

Laboratory 5 (100 Points)

Assignment 1

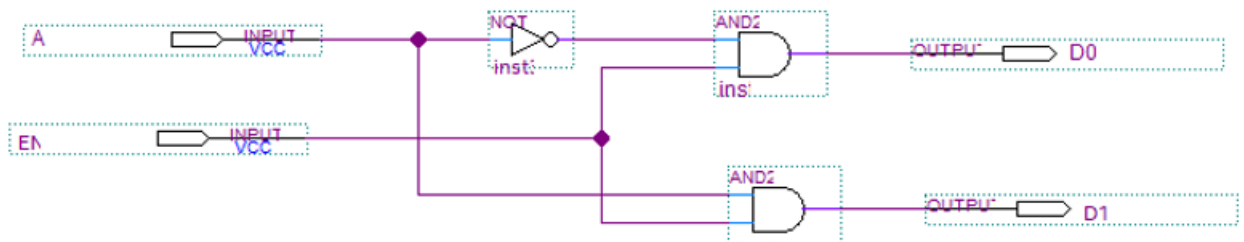
Design a logic diagram to implement 1-2 decoder with Enable pin

1. Derive the decoder's truth table.
2. Design the logic diagram to implement the 1-2 decoder.
3. Simulate the circuit and check the result with the truth table.

En	A	D1	D0
0	X	X	X
1	0	0	1
1	1	1	0

$$D0 = \underline{En}(1) * A'$$

$$D1 = En(1) * A$$



Assignment 2

Design a logic diagram to implement 2-4 decoder with Enable pin (using buses for grouping lines in the same group) using the 1-2 decoders.

1. Derive the decoder's truth table.
2. Design the logic diagram to implement the 2-4 decoder using the 1-2 decoders.
3. Simulate the circuit and check the result with the truth table.

En	A1	A0	D3	D2	D1	D0
0	X	X	X	X	X	X
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

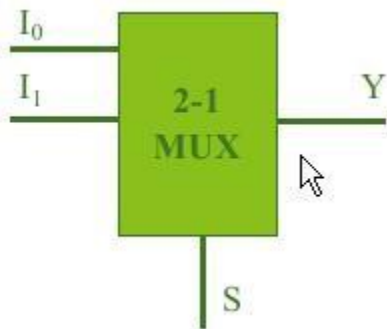
Assignment 3

Design a logic diagram to implement 3-8 decoder with Enable pin (using buses for grouping lines in the same group) using the 1-2 decoders, 2-4 decoders.

1. Derive the decoder's truth table.
2. Design the logic diagram to implement the 2-4 decoder using 1-2 decoders, 2-4 decoders.
3. Simulate the circuit and check the result with the truth table.

Assignment 4

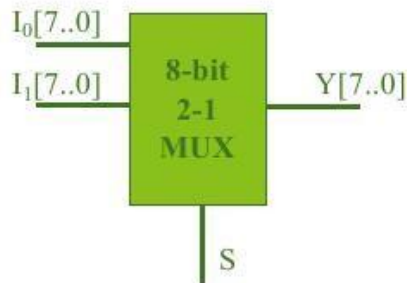
Design a logic diagram to implement 1-bit 2-1 multiplexer



1. Derive 2-1 MUX truth table
2. Design the logic diagram to implement the multiplexer.
3. Simulate the circuit and check the result.

Assignment 5

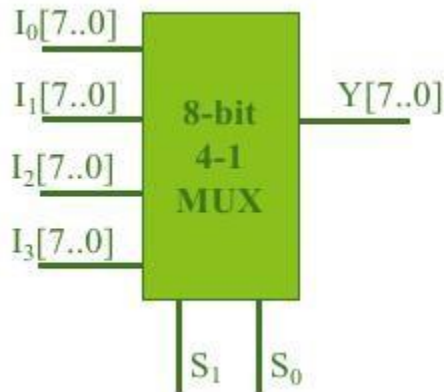
Design a logic diagram to implement 8-bit 2-1 multiplexer



1. Design the logic diagram to implement an 8-bit 2-1 multiplexer using 1-bit 2-1 multiplexers
2. Simulate the circuit and check the result

Assignment 6

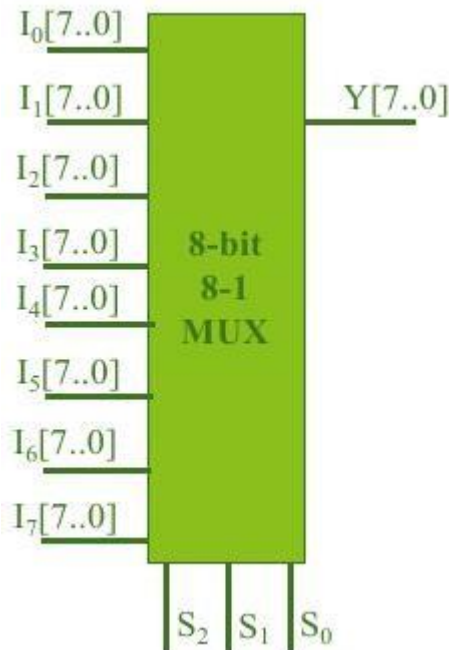
Design a logic diagram to implement 8-bit 4-1 multiplexer using three 8-bit 2-1 multiplexers.



1. Design the logic diagram to implement an 8-bit 4-1 multiplexer using three 8-bit 2-1 multiplexers
2. Simulate the circuit and check the result.

Assignment 7

Design a logic diagram to implement 8-bit 8-1 multiplexer using two 8-bit 4-1 multiplexers and one 8-bit 2-1 multiplexer.



1. Design the logic diagram to implement an 8-bit 4-1 multiplexer using two 8-bit 4-1 multiplexers and one 8-bit 2-1 multiplexer.
2. Simulate the circuit and check the result.

Assignment 8

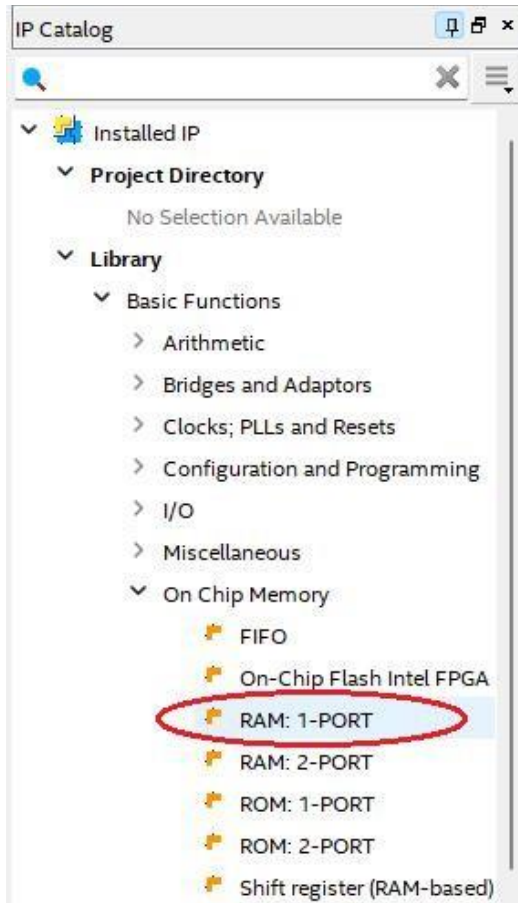
Design a logic diagram to implement an 8-bit parallel load register with enable input.

1. Design the logic diagram of the register.
2. Simulate the circuit and check the result.

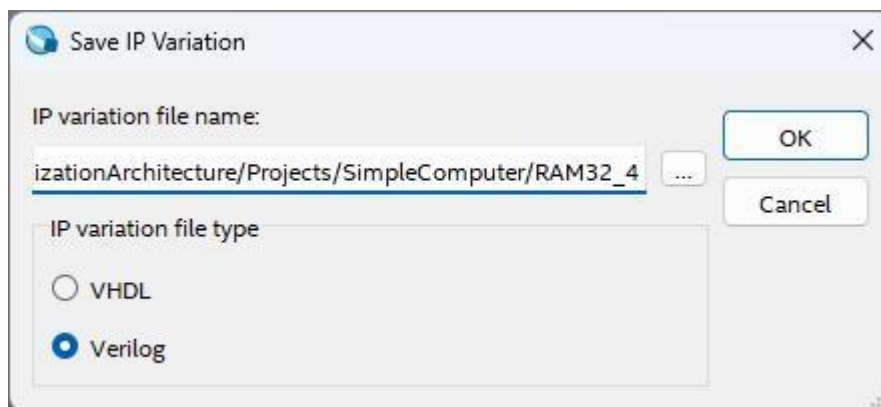
Assignment 9

Using On chip memory RAM: 1-Port, create a 128-bit RAM organized as 32-word by 4-bit array.

1. Create the 32-4 RAM (32 by 4)



- Name the RAM as RAM32_4:



- The **MegaWizard Plug-In Manager** window is shown as in the following screen.

MegaWizard Plug-In Manager [page 1 of 6]

RAM: 1-PORT

AboutDocumentation

1Parameter Settings2EDA3Summary

Widths/Blk Type/Cks>Regs/Cken/Byte Enable/Adrs>Read During Write Option>Mem Init>

RAM32_4

data[3..0]

wren

address[4..0]

clock

4 bits
32 words

q[3..0]

Block type: AUTO

Resource Usage

1 M9K

Currently selected device family: MAX 10

☒ Match project/default

How wide should the 'q' output bus be?

4bits

How many 4-bit words of memory?

32words

Note: You could enter arbitrary values for width and depth

What should the memory block type be?

☒ Auto☐ MLAB☐ M9K

☐ M144K☐ LCsOptions...

Set the maximum block depth to Autowords

What clocking method would you like to use?

☒ Single clock☐ Dual clock: use separate 'input' and 'output' clocks

Cancel< BackNext >Finish

MegaWizard Plug-In Manager [page 2 of 6]

RAM: 1-PORT

About Documentation

1 Parameter Settings 2 EDA 3 Summary

Widths/Blk Type/Cls > Regs/Clen/Byte Enable/Adrs > Read During Write Option

RAM32_4

Block type: AUTO

Which ports should be registered?

- ☒ 'data' and 'wren' input ports
- ☒ 'address' input port
- ☒ 'q' output port

☐ Create one clock enable signal for each clock signal.

☐ Note: All registered ports are controlled by the enable signal(s) [More Options...](#)

☐ Create byte enable for port A

What is the width of a byte for byte enables? 8 bits

☐ Create an 'adr' asynchronous clear for the registered ports [More Options...](#)

☐ Create a 'rden' read enable signal

Resource Usage

1 M9K

Cancel < Back Next > Finish

Notice that the 'q' output port as well as all of the input ports are registered. This means that any data at these ports will be read in on the next rising edge of the clock.



RAM: 1-PORT

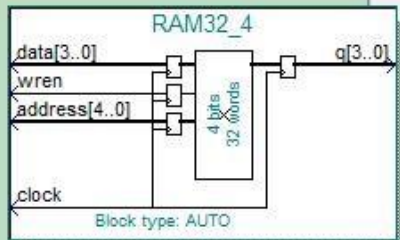
[About](#)[Documentation](#)**1** Parameter Settings**2** EDA**3** Summary

Widths/Blk Type/Cls

Regs/Clsen/Byte Enable/Adrs

Read During Write Option

Mem Init



Single Port Read-During-Write Option

What should the q output be when reading from a memory location being written to?

New Data

- ☒ Get x's for write masked bytes instead of old data when byte enable is used

Resource Usage

1 M9K

Cancel

< Back

Next >

Finish

MegaWizard Plug-In Manager [page 4 of 6]



RAM: 1-PORT

AboutDocumentation

1Parameter Settings2EDA3Summary

Widths/Blk Type/Clocks>Regs/Clock/Byte Enable/Address>Read During Write Option>Mem Init>

RAM32_4

data[3..0]

wren

address[4..0]

clock

4 bits
32 words

q[3..0]

Block type: AUTO

Resource Usage

1 M9K

Do you want to specify the initial content of the memory?

☒ No, leave it blank

☐ Initialize memory content data to XX..X on power-up in simulation

☐ Yes, use this file for the memory content data
(You can use a Hexadecimal (Intel-format) File [.hex] or a Memory Initialization File [.mif])

Note: The configuration scheme of your device is Internal Configuration. In order to use memory initialization, you must select a single image configuration mode with memory initialization, for example the Single Compressed Image with Memory Initialization option. You can set the configuration mode on the Configuration page of the Device and Pin Options dialog box.

Browse...

File name:

The initial content file should conform to which port's dimensions? PORT_A

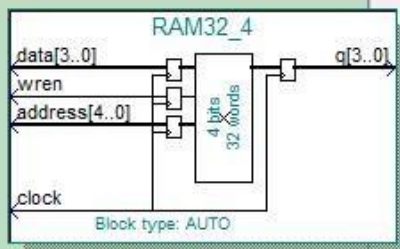
☐ Allow In-System Memory Content Editor to capture and update content independently of the system clock
The 'Instance ID' of this RAM is: NONE

Cancel< BackNext >Finish

We can initialize the memory contents of the RAM later.



RAM: 1-PORT

[About](#)[Documentation](#)**1** Parameter
Settings**2** EDA**3** Summary

Resource Usage

1 M9K

Simulation Libraries

To properly simulate the generated design files, the following simulation model file(s) are needed

File	Description
altera_mf	Altera megafunction simulation library

Timing and resource estimation

Generates a netlist for timing and resource estimation for this megafunction. If you are synthesizing your design with a third-party EDA synthesis tool, using a timing and resource estimation netlist can allow for better design optimization.

Not all third-party synthesis tools support this feature - check with the tool vendor for complete support information.

Note: Netlist generation can be a time-intensive process. The size of the design and the speed of your system affect the time it takes for netlist generation to complete.

☐ Generate netlist

Cancel

< Back

Next >

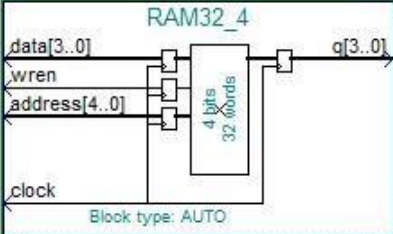
Finish

MegaWizard Plug-In Manager [page 6 of 6]

RAM: 1-PORT

About Documentation

1 Parameter Settings 2 EDA 3 Summary



Turn on the files you wish to generate. A gray checkmark indicates a file that is automatically generated, and a green checkmark indicates an optional file. Click Finish to generate the selected files. The state of each checkbox is maintained in subsequent MegaWizard Plug-In Manager sessions.

The MegaWizard Plug-In Manager creates the selected files in the following directory:
D:\Teaching_labs\ComputerOrganizationArchitecture\Projects\SimpleComputer\

File	Description
☒ RAM32_4.v	Variation file
☐ RAM32_4.inc	AHDL Include file
☐ RAM32_4.cmp	VHDL component declaration file
☒ RAM32_4.bsf	Quartus Prime symbol file
☐ RAM32_4_inst.v	Instantiation template file
☒ RAM32_4_bb.v	Verilog HDL black-box file

Resource Usage
1 M9K

Cancel < Back Next > Finish

Quartus Prime IP Files

When you create an Intel IP variation, a Quartus Prime IP File is generated. Quartus Prime IP Files are used to represent the Intel IP in your design. Do you want to add the Quartus Prime IP File to the project?

☒ D:\Teaching_labs\ComputerOrganizationArchitecture\Projects\SimpleCo...

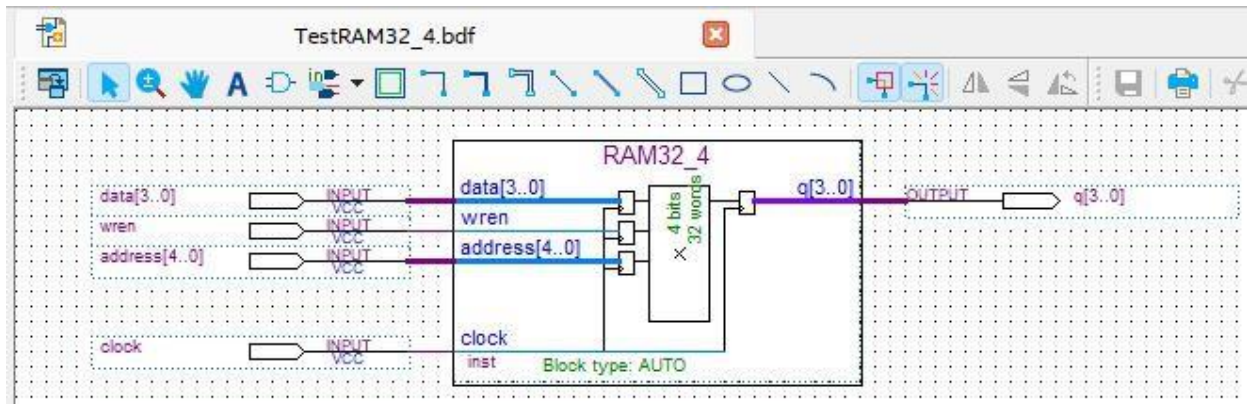
☐ Automatically add Quartus Prime IP Files to all projects

(Note: Turning on this option permanently suppresses this dialog box. You can change this setting in the Options dialog box)

Yes No Help

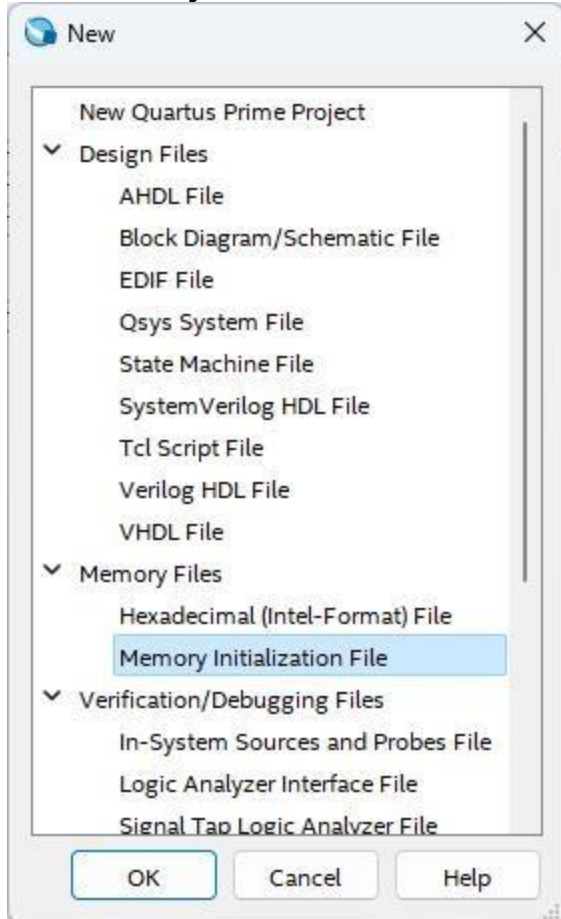
You should now see the symbol for the RAM we just created. Click **Yes** and place the RAM in your design.

2. Use the RAM32_4

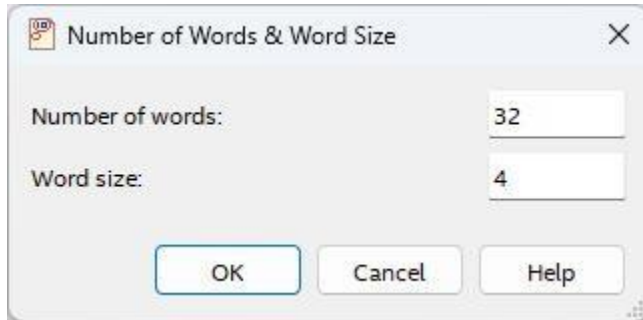


3. We now initialize the memory contents of our RAM.

We will use a **Memory Initialization File (.MIF)**. Select **File > New** from the menu and then select **Memory Initialization File**. Click **OK**.



Select 16 words and 4 bit word size to match our RAM block.



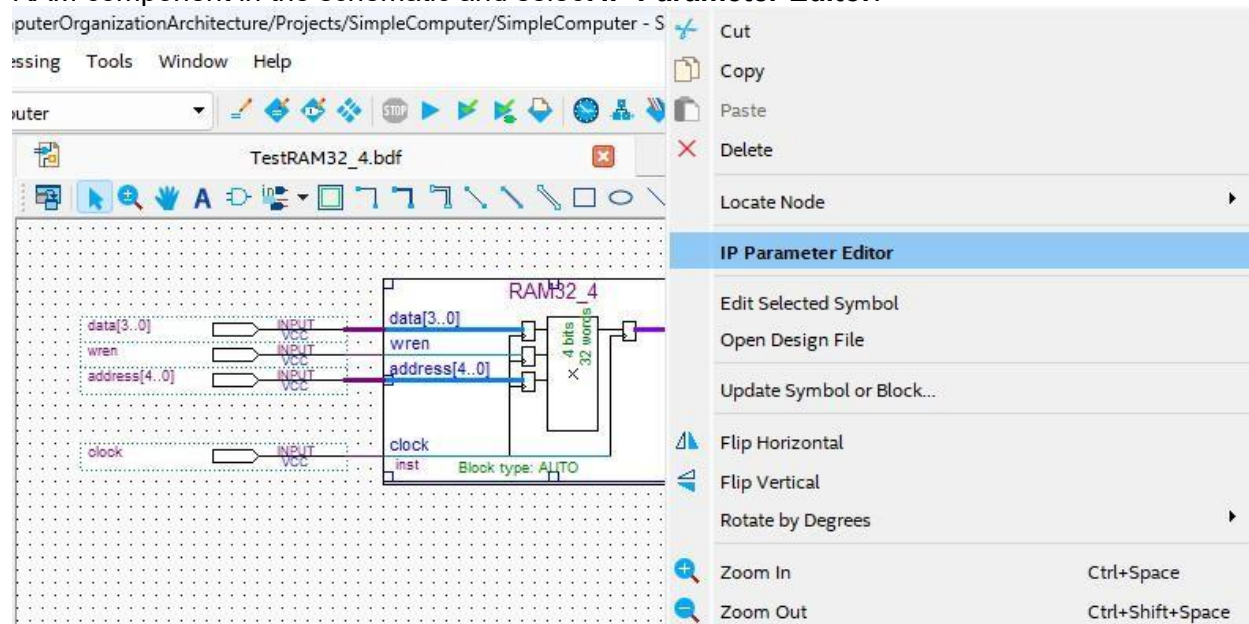
You can now fill the memory with any values from 0 to 15 (15 is the maximum because our word size is 4 bits).

Right clicking on a cell or range of cells will allow us to rapidly fill multiple memory addresses with custom values.

Remember to save the .imf file. The file name should default to RAM32_4_test.mif

Addr	+0	+1	+2	+3	+4	+5	+6	+7	ASCII
0	0	1	2	3	4	5	6	7	
8	8	0	0	0	0	0	0	0	
16	0	0	0	0	0	0	0	0	
24	0	0	0	0	0	0	0	0	

Now we must update our RAM to initialize itself to our memory initialization file. Right click the RAM component in the schematic and select **IP Parameter Editor**.



Click Next until you reach the **Mem Init** tab (or just click the **Mem Init** tab). Select **Yes, use this file for the memory content data** to specify a file and then select browse and find your .mif file. Select the file and then click Finish.

MegaWizard Plug-In Manager [page 4 of 6]

RAM: 1-PORT

About Documentation

1 Parameter Settings 2 EDA 3 Summary

Widths/Blk Type/Cls > Regs/Clen/Byte Enable/Adrs > Read During Write Option > **Mem Init** >

RAM32_4

Block type: AUTO

Do you want to specify the initial content of the memory?

☐ No, leave it blank

☐ Initialize memory content data to XX..X on power-up in simulation

☒ **Yes, use this file for the memory content data**

(You can use a Hexadecimal (Intel-format) File [.hex] or a Memory Initialization File [.mif])

Note: The configuration scheme of your device is Internal Configuration. In order to use memory initialization, you must select a single image configuration mode with memory initialization, for example the Single Compressed Image with Memory Initialization option. You can set the configuration mode on the Configuration page of the Device and Pin Options dialog box.

Browse...

File name:

The initial content file should conform to which port's dimensions?

☐ Allow In-System Memory Content Editor to capture and update content independently of the system clock

The 'Instance ID' of this RAM is:

Resource Usage

1 M9K

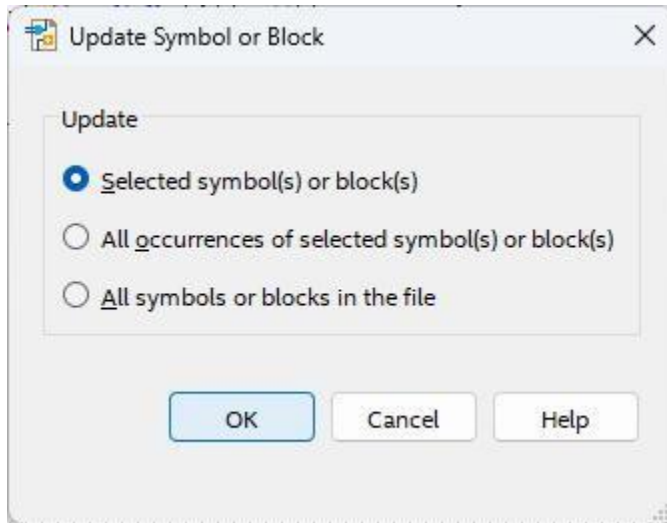
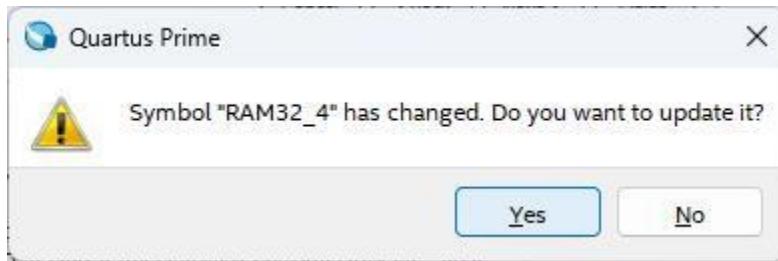
Cancel < Back Next > Finish

MegaWizard Plug-In Manager

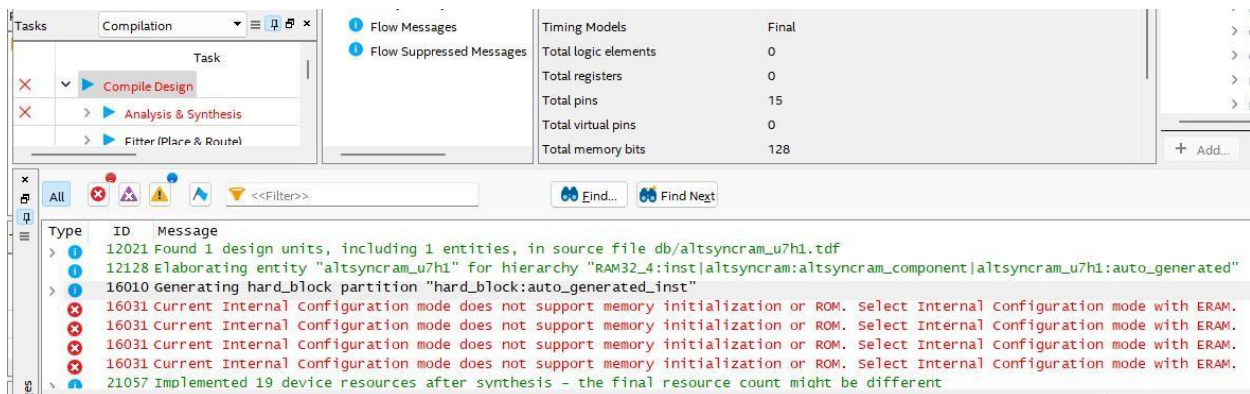
The MegaWizard Plug-In Manager will overwrite the following existing file(s):

- D:/Teaching_labs/ComputerOrganizationArchitecture/Projects/SimpleComputer/RAM32_4.v
- D:/Teaching_labs/ComputerOrganizationArchitecture/Projects/SimpleComputer/RAM32_4.bsf
- D:/Teaching_labs/ComputerOrganizationArchitecture/Projects/SimpleComputer/RAM32_4_bb.v

OK



Compile the design



During compilation if you get the error message:
 "16031 Current Internal Configuration mode does not support memory initialization or ROM. Select Internal Configuration mode with ERAM.",

Select **Assignments -> Device -> Device and Pin Options -> Configuration -> Configuration Mode: Single uncompressed image with Memory Initialization**

Device

Board

Select the family and device you want to target for compilation.

You can install additional device support with the Install Devices command on the Tools menu.

To determine the version of the Quartus Prime software in which your target device is supported, refer to the [Device Support List](#) webpage.

Device family

Family: MAX 10 (DA/DF/DC/SA/SC)

Device: All

Target device

☐ Auto device selected by the Fitter

☒ Specific device selected in 'Available devices' list

☐ Other: n/a

Show in 'Available devices' list

Package: Any

Pin count: Any

Core speed grade: Any

Name filter:

☒ Show advanced devices

Device and Pin Options...

Available devices:

Name	Core Voltage	LEs	Total I/Os	GPIOs	Memory Bits	Embedded multiplier
10M08DAF256A7G	1.2V	8064	178	178	387072	48
10M08DAF256C7G	1.2V	8064	178	178	387072	48
10M08DAF256C8G	1.2V	8064	178	178	387072	48
10M08DAF256C8GES	1.2V	8064	178	178	387072	48
-----	----	----	----	----	-----	--

Migration Devices...

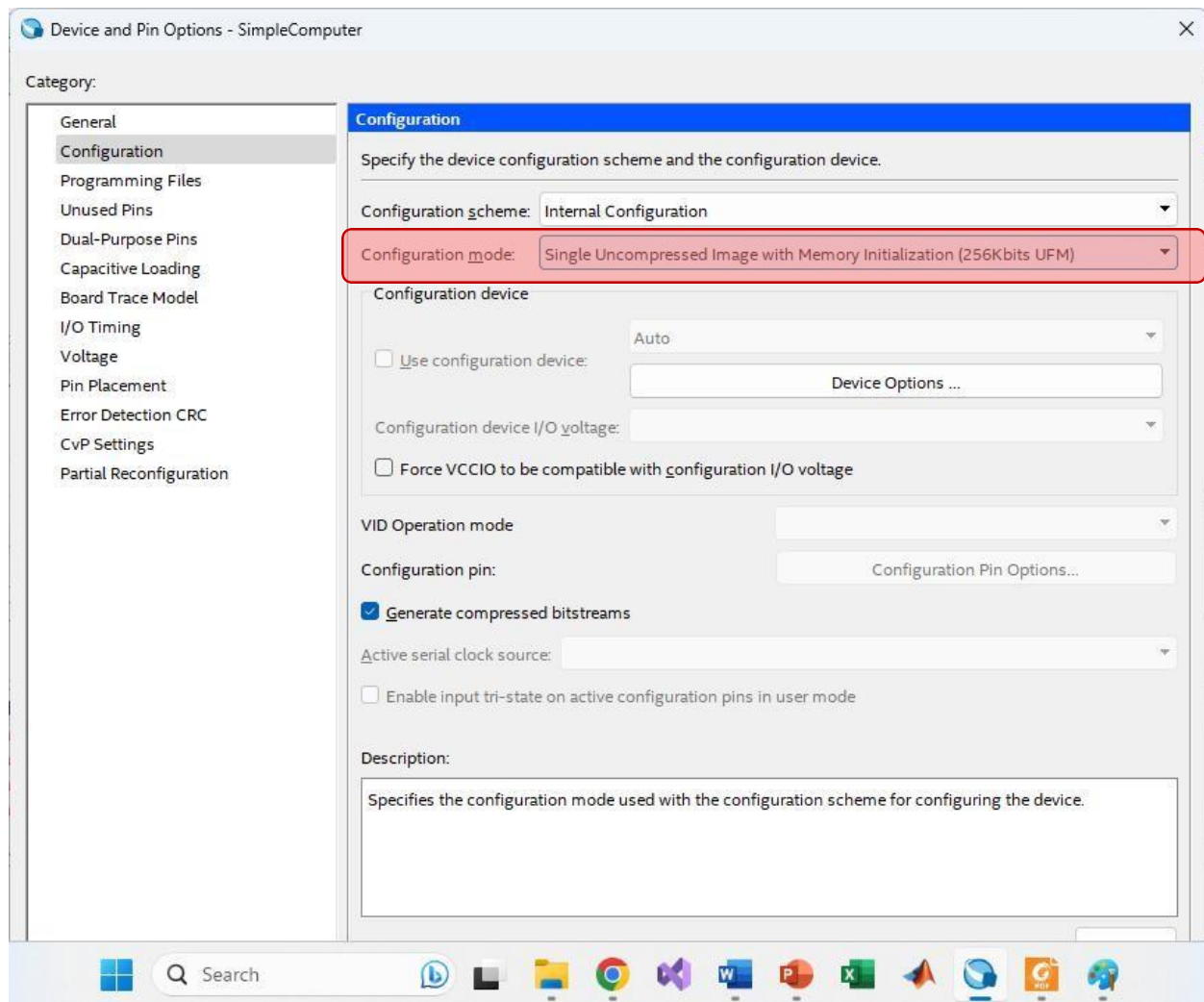
0 migration devices selected

Buy Software

OK

Cancel

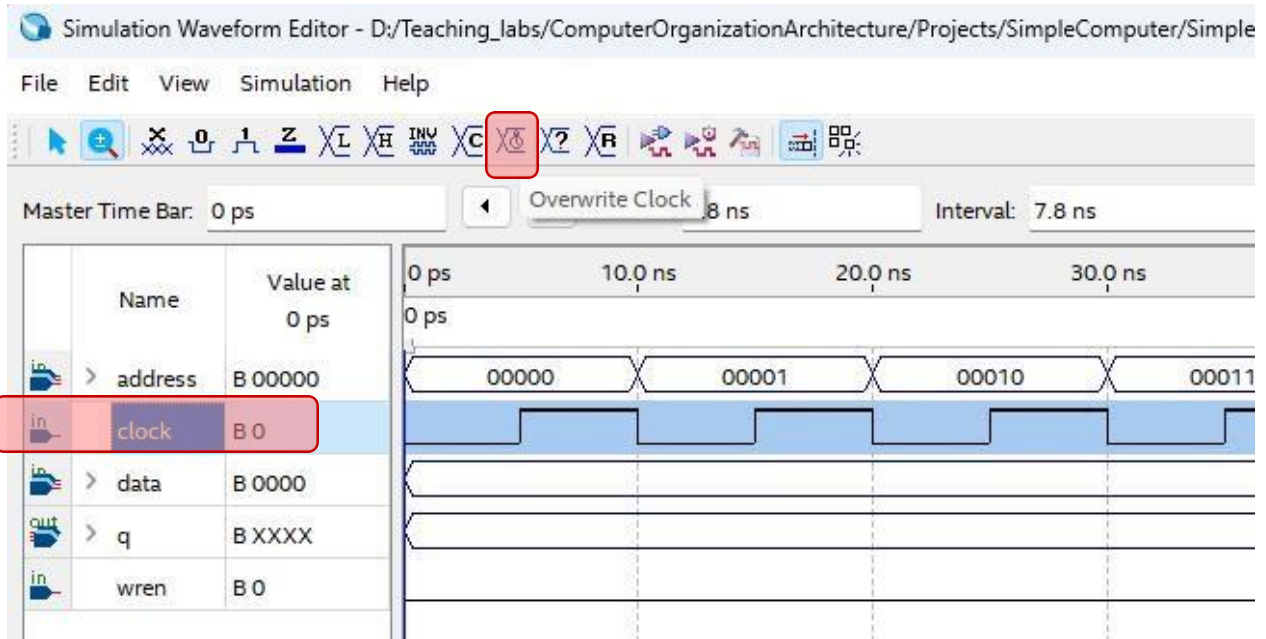
Help



Recompile the project again.

4. Simulate the circuit

- Create the clock signal for the **clock** line:
 - + Select the **clock** line and select **Overwrite Clock**.



We can determine the **Period**, **Offset** and **Duty cycle** of the clock signal.

Clock

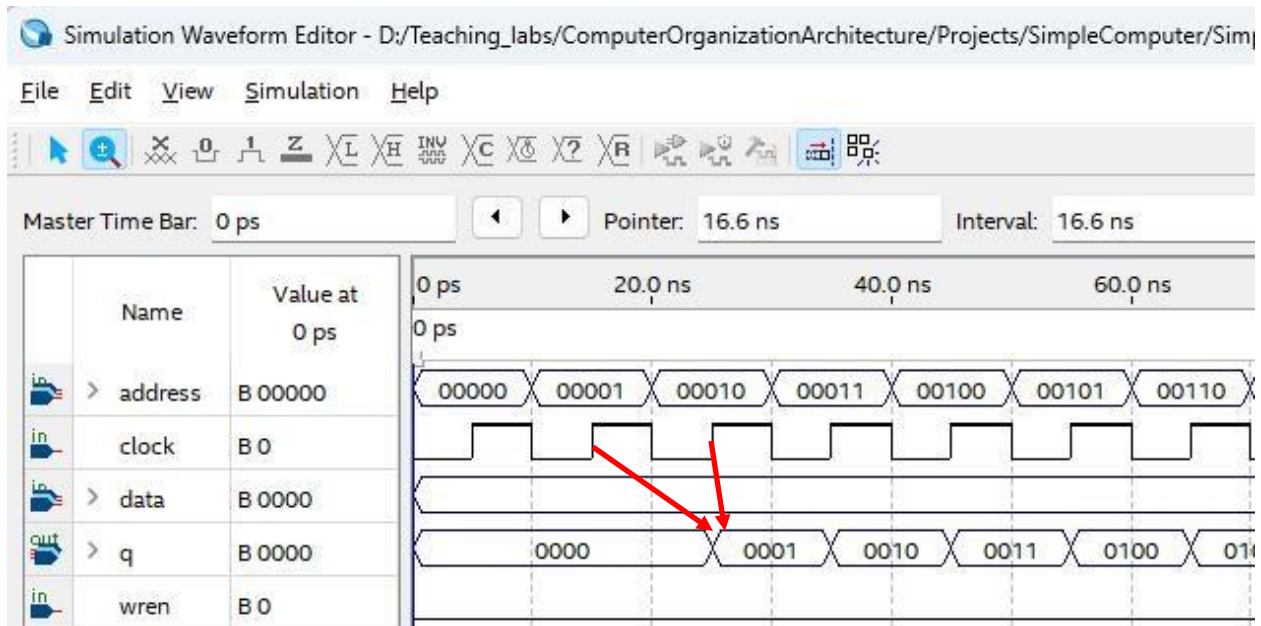
Base waveform on time period

Period: 10.0 ns

Offset: 0.0 ns

Duty cycle (%): 50

OK Cancel



You should see that the **q** output matches the data values we stored in the RAM using our memory initialization file.

Notice that the output is updated on the second rising edge of the clock after a new address is presented to the RAM. This is because we selected to register both the input and output ports of our RAM in the MegaWizard Plug-In Manager. On the first rising edge, the RAM receives whatever address is presented at the address pins. The RAM reads the data value and presents it to the output port, but the data at the output port is not presented to the **q** pins until the next rising clock edge.

Assignment 10

Design a logic diagram of a 32 x 8 RAM using two 32 x 4 RAM chips.

1. Design the logic diagram.
2. Simulate the circuit and check the result.

Assignment 11

Design a logic diagram of a 64 x 8 RAM using two 32 x 8 RAM chips.

1. Design the logic diagram.
2. Simulate the circuit and check the result.